



Amazon Product Review Sentiment Analysis



Project Overview

This project performs **Exploratory Data Analysis (EDA)** on an Amazon Product Reviews dataset to understand customer feedback and product performance. The analysis includes data cleaning, duplicate removal, review length analysis, rating distribution, and visualization of product ratings.

The objective is to discover meaningful insights from customer reviews that can help businesses understand customer satisfaction and improve their products.



Objectives

- Load and explore the Amazon product review dataset.
 - Clean the dataset by removing duplicate records.
 - Check for missing values and understand the dataset structure.
 - Analyze customer ratings.
 - Calculate review lengths.
 - Visualize rating distributions.
 - Identify top-rated products.
 - Generate business insights from customer reviews.
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Technologies Used

Technology	Purpose
Python 3.x	Programming Language
Pandas	Data loading, cleaning, and analysis
NumPy	Numerical computations
Matplotlib	Data visualization
Seaborn	Statistical data visualization
Jupyter Notebook / Google Colab	Development Environment

Dataset Information

The dataset contains Amazon product reviews with information such as:

- Product Name
 - Review Content
 - Rating
 - Review Title
 - Reviewer Details
 - Other product-related attributes
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Project Workflow

1. Data Loading

- Imported the dataset using **Pandas**.
 - Loaded the CSV file into a DataFrame.
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2. Dataset Exploration

Performed basic inspection using:

- Dataset Shape
- Dataset Information
- Column Names
- Descriptive Statistics

This helps understand the size, structure, and data types of the dataset.

3. Data Cleaning

The following preprocessing steps were performed:

- Checked for missing values.
- Identified duplicate records.
- Removed duplicate entries.
- Verified cleaned dataset.

Clean data ensures more accurate analysis.

4. Feature Engineering

A new feature called **Review Length** was created by calculating the number of characters in each review.

This helps analyze:

- Short reviews
 - Long reviews
 - Customer engagement
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5. Exploratory Data Analysis (EDA)

Several analyses were performed including:

Rating Distribution

- Distribution of customer ratings.
- Understanding customer satisfaction levels.



Review Length Analysis

- Distribution of review lengths.
- Helps identify detailed vs short reviews.



Average Rating by Product

Calculated average rating for each product and visualized the results using bar charts.



Visualizations

The project includes several visualizations such as:

- Rating Distribution
- Average Rating by Product
- Review Length Distribution
- Statistical Summary Charts

These visualizations help understand customer behavior and product performance.



Key Insights

- Most products received positive customer ratings.

- Duplicate records were successfully removed.
 - Review lengths vary significantly among customers.
 - Certain products consistently receive higher average ratings.
 - Review length can provide additional context about customer satisfaction.
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Project Structure

```
Amazon-Product-Review-Sentiment-Analysis/  
|  
├─ Amazon_Product_Review_Sentiment_Analysis.ipynb  
├─ amazon.csv  
├─ README.md  
└─ images/  
    ├─ rating_distribution.png  
    ├─ average_rating.png  
    └─ review_length_distribution.png
```

How to Run the Project

1. Clone this repository.

```
git clone https://github.com/your-username/Amazon-Product-Review-Sentiment-Analysis.git
```

1. Navigate to the project folder.

```
cd Amazon-Product-Review-Sentiment-Analysis
```

1. Install the required libraries.

```
pip install pandas numpy matplotlib seaborn
```

1. Open the notebook using Jupyter Notebook or Google Colab.
 2. Run all cells sequentially.
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Skills Demonstrated

- Data Cleaning
 - Exploratory Data Analysis (EDA)
 - Feature Engineering
 - Data Visualization
 - Business Insight Generation
 - Python Programming
 - Pandas
 - NumPy
 - Matplotlib
 - Seaborn
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Future Improvements

- Perform Natural Language Processing (NLP).
 - Apply sentiment classification using machine learning.
 - Build predictive models for customer sentiment.
 - Develop an interactive dashboard using Streamlit or Power BI.
 - Deploy the project as a web application.
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Conclusion

This project demonstrates the complete workflow of cleaning, analyzing, and visualizing Amazon product review data. It highlights how Python and data analysis libraries can be used to transform raw customer feedback into actionable business insights. The project serves as a strong foundation for future sentiment analysis and machine learning applications.



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